Xin Ma

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EDUCATION

2022 – Present **University of Florida**, Gainesville, FL, USA

- Ph.D. in Biostatistics
- Advisors: Li Chen, Ramon C. Sun
- Expected graduation: Summer 2027

2018 – 2020 **The George Washington University**, Washington, D.C., USA

- M.S. in Statistics

2014 – 2018 **Southwestern University of Finance and Economics**, Chengdu, China

- B.S. in Actuarial Science

RESEARCH INTERESTS

Methodology Spatial and Graph-based Modeling; Statistical Machine Learning; Multimodal Data Integration; Interpretable AI.

Applications Spatial Multi-omics; Single-cell Multi-omics; Biomedical Imaging; Neurodegenerative Diseases; Cancer.

PUBLICATIONS

† indicates co-first author.

Peer-Reviewed Publications

1. **Ma, X.**, Allison, D.B., Macedo, J.K.A., Bolton, K.E., Clarke, H.A., Young, L.E.A., Bian, J., Neltner, J.H., Vander Kooi, C.W., Sun, R.C., Gentry, M.S., and Chen, L. *Machine-learning-based prediction of clinical outcomes in gliomas using glycomic features.* ***Computational and Structural Biotechnology Journal*** (2026).

2. **Ma, X.**, Jin, W., Lu, Q., Sun, R.C., and Chen, L. *Reference-informed spatial domain detection using weak supervision for spatial transcriptomics.* ***Genome Research***, 36(7), 1466–1482 (2026).

3. Hawkinson, T.R.†, Liu, Z.†, Ribas, R.A., Medina, T., Nielsen, R.S., Clarke, H.A., **Ma, X.**, Mueller, A.C., Plasencia, A.F., Sheer, A.L., Simpson, S.T., Soto, C.M., Sudderth, J., Cai, F., Cantrell, A.R., Colpaert, M.G., Shedlock, C.J., Wu, L., Young, L.E.A., Kooser, D.D., Chen, L., Ryan, A.M., Quinones, S., Son, J., Azadi, P., DeBerardinis, R.J., Prokop, S., Allison, D.B., Yang, S., Chen, H., Huang, Y., He, X., Alonge, K.M., Guo, J., Guo, Y., Bian, J., Vander Kooi, C.W., Gentry, M.S., and Sun, R.C. *Hyperglycosylation is a metabolic driver of Alzheimer's disease.* ***Nature Metabolism***, 8, 1410–1425 (2026).

4. Ribas, R.A., Tang, Q., Caffee, S.I., Shedlock, C.J., Hawkinson, T.R., Dragich, A.K., Ziani, B.E.C., Bucco Paolasso, F., Black, A.P., **Ma, X.**, Kota, H.C., Chen, L., Frolenkov, G.I., Allison, D.B.,

Gentry, M.S., Sun, R.C., and Vander Kooi, C.W. *Mass spectrometry-based spatial imaging of the cochlea.* ***Journal of the American Society for Mass Spectrometry***, 37(5), 1187–1195 (2026).

5. Clarke, H.A.†, **Ma, X.**†, Shedlock, C.J.†, Medina, T., Hawkinson, T.R., Wu, L., Ribas, R.A., Keohane, S., Ravi, S., Bizon, J.L., Burke, S.N., Abisambra, J.F., Merritt, M.E., Prentice, B.M., Vander Kooi, C.W., Gentry, M.S., Chen, L., and Sun, R.C. *Spatial mapping of the brain metabolome lipidome and glycome.* ***Nature Communications***, 16, 4373 (2025).

6. **Ma, X.**†, Shedlock, C.J.†, Medina, T.†, Ribas, R.A.†, Clarke, H.A., Hawkinson, T.R., Dande, P.K., Golamari, H.K.R., Wu, L., Ziani, B.E.C., Burke, S.N., Merritt, M.E., Vander Kooi, C.W., Gentry, M.S., Yadav, N.N., Chen, L., and Sun, R.C. *AI-driven framework to map the brain metabolome in three dimensions.* ***Nature Metabolism***, 7, 842–853 (2025).

7. Clarke, H.A.†, Hawkinson, T.R.†, Shedlock, C.J., Medina, T., Ribas, R.A., Wu, L., Liu, Z., **Ma, X.**, Xia, Y., Huang, Y., He, X., Chang, J.E., Young, L.E.A., Juras, J.A., Buoncristiani, M.D., James, A.N., Rushin, A., Merritt, M.E., Mestas, A., Lamb, J.F., Manauis, E.C., Austin, G.L., Chen, L., Singh, P.K., Bian, J., Vander Kooi, C.W., Evers, B.M., Brainson, C.F., Allison, D.B., Gentry, M.S., and Sun, R.C. *Glycogen drives tumour initiation and progression in lung adenocarcinoma.* ***Nature Metabolism***, 7, 952–965 (2025).

8. **Ma, X.**, Thela, S.R., Zhao, F., Yao, B., Wen, Z., Jin, P., Zhao, J., and Chen, L. *Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.* ***Bioinformatics***, 40(9), btae528 (2024).

9. Zhao, F., **Ma, X.**, Yao, B., Lu, Q., and Chen, L. *scaDA: A novel statistical method for differential analysis of single-cell chromatin accessibility sequencing data.* ***PLOS Computational Biology***, 20(8) (2024).

Manuscripts Under Revision

1. Rodriguez-Palma, E., Gomez, K., Calderon-Rivera, A., et al., including **Ma, X.** *A peptidomimetic inhibitor of moesin alleviates chronic pain by suppressing NaV1.7 and NaV1.8 channels.* ***Science Translational Medicine*** (under revision).

2. Guarnieri, J.W., Maghsoudi, Z., Kim, J., et al., including **Ma, X.** *Guardians of the mitochondria: Space Mitochondria 2.0 systemic analysis reveals bioenergetic dysregulation across species.* ***Cell Press Blue*** (under revision).

3. Jin, W., Zhao, F., **Ma, X.**, Lu, Q., Esser, K., Sun, R.C., and Chen, L. *DeepCNet: A multimodal deep learning model for predicting cell type-specific gene expression and promoter-enhancer interactions from single-cell multiome data.* ***Nature Communications*** (under revision).

4. Liu, Z.†, **Ma, X.**†, Ribas, R.A., Medina, T., Quinones, S., Son, J., Wu, L., Ryan, A.M., Shedlock, C.J., Ziani, B., Barco-Caiaffa, V., Rao, N., Wong, K., Titus, A., Larson, R., Vander Kooi, C.W., Chandel, N.S., Gentry, M.S., Chen, L., and Sun, R.C. *Metabolic coherence of the mouse brain.* ***Cell*** (under revision).

5. Medina, T., Quiñones, S., Liu, Z., Wu, L., Ziani, B., Shedlock, C.J., **Ma, X.**, Ryan, A.M., Larson, R., Soto, C.M., Barco-Caiaffa, V., Titus, A., Wong, K., Rao, N., Florea, S., Gutierrez-Monreal, M.A., Mitchell, G.S., Vander Kooi, C.W., Gentry, M.S., Chandel, N.S., Esser, K.A., and Sun, R.C. *A multi-organ spatial metabolomic atlas of exercising mice reveals neuronal Complex I as a convergent and sufficient axis for tau pathology reduction in PS19.* ***Science*** (under revision).

TALKS AND PRESENTATIONS

Oral Presentations

2026 UF AI Days, Gainesville, FL.
From 2D to 3D: AI-Driven Mapping of Brain Metabolism and Its Implications for Alzheimer's Disease.

2026 ENAR Spring Meeting, Indianapolis, IN.
Reference-informed spatial domain detection using weak supervision for spatial transcriptomics.

2024 UF PHHP Research Days, Gainesville, FL.
Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.

2024 Annual ASA Florida Chapter Meeting, Tallahassee, FL.
Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.

2023 Mark C. K. Yang Event, Gainesville, FL.
Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.

2023 International Conference on Intelligent Biology and Medicine (ICIBM), Tampa, FL.
Spatial mapping of the brain metabolome lipidome and glycome.

Poster Presentations

2025 ENAR Spring Meeting, New Orleans, LA.
Reference-informed spatial domain detection using weak supervision for spatial transcriptomics.

2025 Winter Conference on Brain Research (WCBR), Lake Tahoe, CA.
AI-driven framework to map the brain metabolome in three dimensions.

2024 MidSouth Conference on Computational Biology and Bioinformatics (MCBIOS), Atlanta, GA.
Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.

2023 Florida Genetics Symposium, Gainesville, FL.
Deep5hmC: predicting genome-wide 5-hydroxymethylcytosine landscape via a multimodal deep learning model.

HONOR AND AWARDS

2026 Biostatistics PhD Student Travel Award
University of Florida.

2024 MCBIOS Travel Fellowship
MidSouth Conference on Computational Biology and Bioinformatics.

2023 Mark C. K. Yang Student Presentation Award
University of Florida.

2018–2019 Dean's Fellowship Award
The George Washington University.

2018 Outstanding Thesis Award
Southwestern University of Finance and Economics.

2014–2018 Outstanding Academic Performance Award
Southwestern University of Finance and Economics.

2015, 2016 Outstanding Social Practice Award
Southwestern University of Finance and Economics.

CONSULTING EXPERIENCE

2023 - Present **Center for Advanced Spatial Biomolecule Research (CASBR)**, University of Florida, Gainesville, FL, USA
Bioinformatics Specialist

- Provide bioinformatics, statistical, and machine-learning consultation for multidisciplinary spatial biology studies.
- Design and implement computational workflows for spatial metabolomics, spatial transcriptomics, mass spectrometry imaging, and multi-omics data integration.
- Advise collaborators on experimental design, data processing, statistical analysis, visualization, biological interpretation, and reproducible research practices.
- Contribute computational analyses and methodological expertise to peer-reviewed publications and manuscripts spanning neuroscience, cancer, and other biomedical applications.

PROFESSIONAL EXPERIENCE

2026 **Eli Lilly and Company**, Indianapolis, IN, USA
PhD AI Discovery Data Science Intern

- Developed a sequence- and graph-based deep learning model to predict siRNA off-target effects, built on a large-scale internal dataset curated from differential-expression experiments
- Improved AUPRC 3x and AUROC 24% over an internal seed-match baseline, and identified a "non-specific responder" gene signature explaining which target genes the model could predict most reliably (Spearman ρ=0.75)
- Diagnosed key data and modeling limitations, validated model robustness on a curated toxicity-relevant gene subset, and presented findings and recommendations to cross-functional stakeholders

2020 – 2022 **TechData Service Company**, King of Prussia, PA, USA
Statistical Programmer and Database Developer

- Conducted statistical analysis for ∼20 Phase I–III clinical trials using SAS; produced TLFs for safety and efficacy.
- Developed and validated CDISC SDTM/ADaM datasets; ensured regulatory-compliant data standards.
- Built and maintained EDC database builds (IBM Clinical Development), including edit checks and CRF modules.
- Drafted and revised Statistical Analysis Plans (SAPs) for multiple clinical trials.

PROFESSIONAL SERVICE

Session Chair: ENAR 2026
Journal Reviewer: PLOS ONE, Cell & Bioscience.

TEACHING

Teaching Assistant

Fall 2022	PHC 6089: Public Health Computing.
Fall 2022	PHC 6050: Statistical Methods for Health Science I.

MEMBERSHIPS

American Statistical Association (ASA)
Eastern North American Region of the International Biometric Society (ENAR)
MidSouth Computational Biology and Bioinformatics Society (MCBIOS)
International Conference on Intelligent Biology and Medicine (ICIBM)

SKILLS

Programming: Python, R, MATLAB, SAS, SQL, Linux Shell, LaTeX.
AI & Machine Learning: PyTorch, RAPIDS.
Computing: HPC, CUDA, Git, Docker.
Languages: Chinese (native), English (fluent).